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RESEARCH PAPER

Adaptive Model Predictive Control for Solar Thermal Systems Under Dynamic Conditions

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Abstract

This study explores an adaptive forward-looking control method for solar heat energy systems, focusing on improving short-term recovery and system output during changing sunlight levels. Current control ways use mostly fixed models or simple gain–reset logic, which fail to keep steady heat levels and actuator working ability during changing outside weather. These ways often react slowly to sunlight shifts, causing energy waste and more damage to moving parts. To solve these weak points, a flexible model-based prediction control plan is shared. The hard part is to keep a good mix of forecast quality, fast system reaction, and part safety without causing delays in running the control tool. Solving this hard part is important, as steady solar heat control helps directly in saving energy and system health. The study aims to build and test a control setup that can react before sunlight changes happen. A MATLAB/Simulink-based setup is used, adding sunlight prediction data and heat flow features. Main system checks include heat difference, size of valve action, return time, and heat use rate—picked based on how the system reacts to sunlight shifts. Test data is made by fixed-case runs of sunlight drop events and studied using signal test plots, control signal shape checks, and sun change-based control steps. System checks focus on lowering heat change, lowering control signal force, and raising energy output. When compared with the initial standard models, the results show strong gains in many system checks. To be clear, return time drops by more than heat use goes up by close to optimum points, and valve use is cut down by up to huge percentage. While this method works well in test sun-drop cases, later works must focus on system growth, long-term part use, and working with backup energy parts.

Keywords Adaptive control · Solar thermal systems · Energy efficiency · Dynamic conditions · System identification

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1 Introduction

The key problem addressed in this work is the unstable temperature behavior in solar thermal loops under fluctuating irradiance, and solar thermal power generation, mainly through a parabolic trough collector (PTC) system, shows a huge promise for a clean and renewable energy output, and directly connects with COP26 global targets on reducing carbon emissions through efficient renewable energy utilization. However, the working performance of these systems is heavily affected by the differences in the environment, especially by strong changes in the sunlight caused by moving clouds, an air flow difference, and shifting weather conditions. These differences can cause huge changes in the output temperature of the heat transfer fluid (HTF), directly limiting the thermal efficiency, the running safety, and the long-term surety in the solar thermal plants. Standard control methods, often using fixed settings

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or basic Proportional-Integral-Derivative (PID) controllers, fail to handle these fast-changing input conditions, which lead to regular temperature differences, more energy waste, and a faster damage to the machine parts. To fix these problems, better control methods that can expect and manage the changing weather are seriously needed. A Model Predictive Control (MPC) has become a strong method due to its ability to plan in advance and use standard rule-based control decisions. But the regular MPC depends too much on fixed models, which may not fit the real-time changes in the system's working setup. An Adaptive Model Predictive Control (AMPC) improves this method by regularly updating the system model with new and fresh real-time data, allowing a stronger and faster control response. This research focuses on building an AMPC method for PTC solar systems, using real weather input and ongoing system checking to keep HTF temperatures steady and improve the total power output. The new control method works to limit the temperature differences, reduce machine stress, and improve the use of energy, helping the solar thermal plants become more steady and surer.

Recent studies have tried the AMPC methods inside a different energy system. A study has shown its useful effect in a heat pump-aided solar water heating setup, showing a better fluid temperature control. However, this research was fixed only to a small-scale type residential setup and it did not take the huge difference faced in the large-scale solar thermal plant (Zhao et al. 2023). Another investigation has applied an adaptive model predictive control to a thermal behavior of a building, getting some energy saving through a real-time adjusting. Yet, this work focused more on a building-level heating, air flow, and air-cooling systems without looking into a fast unstable behavior that happens in a solar collector loop (Tohidi et al. 2024). Another study has made an adaptive predictive control rule for the heating system under a varying load, but it has missed the joining of a quick sunlight forecasting that is very much needed in the solar thermal setup (Sha et al. 2023). While a model predictive control method was tested under an unsure forecast for a building having a solar energy system, the non-stop real-time model correction needed for the parabolic trough collector was not added (Liu et al. 2018). A study on a residential solar-air joined heat pump system has shown better comfort control, but the huge solar collector space and the outside weather issue changing the steady state fluid temperature were not talked about (Zhao et al. 2024). An evaluation on a predictive controller for the solar thermal heat system with a heat storage under a changing people usage also skipped the fast control action needed for sunlight radiation difference (Wei 2023). Another smart method has focused on a mixed radiant-air cooling tool in the desert-type weather, but it did not keep a focus on a solar collector work that is important for the parabolic trough type system (Hassan and Abdelaziz 2022). In one more field, a model predictive controller was tested for a solar-thermal reactor, showing a better system balance; but the reactor's batch setup is not the same as a constantly moving solar collector loop (Saade et al. 2014). Even though some predictive control rules were made for the latent heat solar thermal systems, the fast-changing features or real-time model change steps were not present, limiting the action in fast sunlight changes (Serale et al. 2018). Some works on a building with a solar-electric heat collector and the active thermal storage used a model predictive control, but missed to stress the need for a quick response time that is so important for the solar field working (Li et al. 2015).

Nonlinear adaptive control was tried for a heat transfer fluid temperature management in parabolic trough solar power plants, though any real-time model refreshes with weather guess data joining were not used, limiting a predictive ability during a fast solar input difference (Nevado Reviriego et al. 2017). Furthermore, an adaptive nonlinear predictive control helped for solar furnace thermal pressure checking, yet a method still stays tied to a testing setup instead of a constant solar field working (Pataro et al. 2024). Adaptive predictive control was also used for photovoltaic panel temperature checking, but this stayed more on photovoltaic units and not on thermal collectors where a heat fluid flow plays a big part (Kumar and Tangirala 2021). A many-model predictive controller for a superheated steam heat in solar parabolic-trough systems showed progress, but it had trouble with using real-time solar guess values and quick model change (Wang et al. 2022). A nonlinear predictive control for a thermal balance in solar trough plants made better system stability, though no adaptive refresh based on live weather values was added (Gallego et al. 2022). Some uses in a thermal energy holding and comfort control inside smart energy grids also showed model predictive control's use; but the unique heat delay and sunlight jump in solar fields need more quick setups (Tang and Wang 2019). Adaptive many-model predictive control took on the unsure nature in building thermal storage but missed doing anything for the solar field changes (Kim 2013). Changed model predictive controllers for top power output checking in photovoltaic units under part shading were studied, but these did not cover a heat transfer fluid control or a big thermal field difference (Siddique et al. 2024). A few studies between a fixed and adaptive linear grey-box model for houses showed better forecast skill, but solar collector systems have harder and more mixed dynamic issues (Yu et al. 2024). A cheap predictive controller was tested on-site for a building heat system with a phase change setup, though this kind of control still needs checking for an outdoor solar heat field with huge changes (Wei and Calautit 2024).

Coordinated predictive scheduling with a real-time adaptive control was used for an integrated building energy setup; however, the direct use in high heat solar collector loops with outdoor change issues stays very limited (Tang et al. 2025). A detailed review on a predictive control for an electric vehicle joining gave helpful structure but did not go for a solar thermal system change (Minchala-Ávila et al. 2025). An informer-based control structure for a district heating showed better heat use planning, yet a fluid heat control under sunlight input jump was not focused (Guo et al. 2024). A heuristic control method helped energy choice in electric school setups, while a solar heat plant needs very quick change to a fast weather jump (Morovat et al. 2024). A quick solar power planning joined with a predictive method fixed some tool waste, but the thermal fluid in hot collectors needs a very exact control setup, and the present work further introduces precise predictive control for thermal fluid along with enhanced operational management not addressed in (Li et al. 2024). A neural control system made building energy use better, though solar collectors meet tough problems from fast input change and fluid heat stay (Agouzoul et al. 2024). Some data-led predictive setups with fair storage energy savings were made, but these did not try to keep heat output fixed during sun change (Sun et al. 2024). A predictive method was also used for cold storage air systems with many goals, but the heat handling is far from the solar collector rise (Zhao et al. 2024). A compare study between fuzzy and predictive controls in solar power top point choice gave learning, but it cannot be used on the thermal fluid line (Li et al. 2025). A proven test shape for predictive control of heat pump helped with big use, while the outdoor solar field needs more model change for outside jump (Mugnini et al. 2024).

A power saving rule was tried on many-home buildings with a predictive method; but the heat move in solar fluid lines needs its own way (Cid et al. 2024). A best load use plan with a nonlinear control in mixed energy sites showed hope, yet the use for a high-heat solar fluid line was not looked at (Li et al. 2024). A money-saving control for a building heater got many compare checks between a model type and a data type, but the fast heat jump in solar tools is harder (Zheng et al. 2024). A two-part rule that joined a predictive and a fixed control made a converter better, but no such way was made for a thermal fluid unit yet (Marmol et al. 2024). A water check idea in a simulator with a predictive and best-fit method made new plans, but this did not solve the real-time solar heat variation in the fluid (Debroy et al. 2025). A two-level planner for a small power grid with a smart random predictive control got a working boost, but the solar fluid loop needs faster auto-back control (Hu et al. 2025). A many-time control for a mixed power grid with a hydrogen save gave smoother results, but the solar fluid needs its own heat fix way (Abdelghany et al. 2024). A many-model controller for a heat inlet in a acid-making line gave good heat hold, though it is not like a solar loop setup (Liu et al. 2024). A gene-based predictive system with a gene rule helped in small grid jump fix, while the live control for solar fluid heat under sky change still needs study (Cavus and Allahham 2024). A smart house power use plan with a live prediction in a solo solar plus battery line got better work, but the heat fluid shift in a solar plant was not taken on (Watil 2024).

Fractional order proportion integral derivative controllers joined with a predictive control helped in a frequency hold for renewable energy groups but did not check a thermal surety under a solar input difference (Gulzar et al. 2024). A learning-led strong predictive method for wave energy tools gave new thoughts, though solar heat fields need a different model fix for the heat fluid steps (Zhang et al. 2024). Mixed predictive plans for electric car heat control made energy better, yet a heat fluid flow in solar loops has quicker change problems (Wang et al. 2024). Better inverter control with adaptive tracking and neural forecast control gave stronger electric tool use, while outside solar heat setups still lack such smart control (Kinga et al. 2024). A smart change in forecast control for a huge wind link gave good advice, but a heat solar unit needs its own fluid control rule (Lu 2024). Some new predictive plans for a grid-helped heat pump gave more use space, but they are not good to be used with solar heat fluid loops (Klepic et al. 2024). A clean energy plan for a multi-area building with a forecast and a machine rule made less carbon use, while a solar field needs extra change to match sun input changes (Chen and You 2024). A control of a solar-battery line helped a weak grid run better but did not cover a high heat solar tool setup (Selim et al. 2024). A power plan for an air cooler load to help grid changes gave more support, but it does not check on a solar heat fluid line heat hold (Zhu et al. 2024). At last, a live change control in a stand-alone small grid with a car gave better grid hold, though a heat stay in a changing solar field is still a problem to work on (Santhi Mary Antony et al. 2024). Recent studies such as (Praveenkumar and Kumar 2025)have explored not only the thermodynamic characteristics but also the environmental impacts together with economic consequences of integrating solar photovoltaic PV panels under various operational and climatic conditions, while providing indirect inspiration together with preliminary groundwork for adaptive control strategies that aim to handle large variations and non-uniformities in solar thermal collection fields especially under fluctuating radiation inputs, and simultaneously in a somewhat related but technically distinct study (Praveenkumar et al. 2023), the researchers examined a hybrid photovoltaic/thermal PV/T system with combined energy output measurement and performance assessment for multiple operating conditions, and Table1 attempts to summarize a total of fifty-two previously published studies related to solar thermal technologies together with adaptive predictive control system frameworks, carefully outlining the main themes together with reported findings and stated limitations while also pointing out significant areas requiring further exploration and detailed technical investigation in future works. Unlike conventional PI/PD controllers and standard MPC frameworks that rely on fixed parameters, the proposed adaptive MPC dynamically updates its internal model using real-time feedback, enabling superior tracking performance and robustness under varying solar thermal conditions.

Though a big set of studies have used a smart forecast control on different heat and energy setups, most work on house types, tiny areas, or a power-only use. A few try to work on a huge solar heat plant, more so on a curved solar collector type with fast sun light change. The known ways often skip live model change, quick event fix, and ongoing model shift for weather moves. Also, only a small set aim at both heat rule fit, system work level, machine use limit, and fast fix to sun jumps. This study tries to close this gap by making a live state space model with a smart change-based forecast control that helps in a heat hold, use boost, and live sun fit under heavily changing sky states sky states. This study tries to close this gap by making a live state space model with a smart change-based forecast control that helps in a heat hold, use boost, and live sun fit under heavily changing sky states change-heavy sky states, and unlike earlier reports this work adds a distinct scientific step by joining live adaptive model shift with forecast-driven input correction inside a large curved solar plant so that the gain is not only practical but also method-wise novel beyond an incremental rise. The aim of this work is to fix some known gaps in a basic solar control by trying five key goals. First, this work sets a live space model to copy the heat and a tool state of a curved solar collector plant. Then, a smart forecast rule is made to follow a live sun shift with a good response. Third, the method tries to hold a heat line steady, so that it gets a tight thermal check. Fourth, the plan wants to boost energy use and limit load on parts. At the end, this setup is checked for its skill to deal with short change times, which may help link it with a heat store in the next work.

The main key innovations and technical contributions of the present research study are described as follows in a detailed and organized manner:

- The research involves the development of a forecast-driven and adaptive Model Predictive Control (AMPC) framework that is specifically designed and suitably tailored for the parabolic trough solar thermal system operating under continuously changing and dynamic irradiance condition difference.
- The study also includes the proper integration of a live and continuously updated state-space model together with a real-time feedback mechanism in order to enhance the thermal tracking performance and to improve the overall actuator efficiency under different operating conditions.
- The validation of the proposed AMPC-based control system is carried out through a detailed simulation-based study using the MATLAB/Simulink software

Table 1 Summary of prior studies in solar thermal and adaptive control systems

Main Focus	Key Findings	Limitations	References
Adaptive MPC for residential solar water heating and heat pumps	Improves energy efficiency, thermal comfort, and response time	Mostly small-scale systems, limited consideration of dynamic solar variability	(Zhao et al. 2023, 2024; Tohidi et al. 2024; Sha et al. 2023; Wei 2023)
MPC under forecast uncertainty for integrated solar systems and buildings	Handles solar irradiance variability and improves system stability	Limited large-scale or industrial-scale validation	(Liu et al. 2018; Li et al. 2015; Tang and Wang 2019; Kim 2013; Yu et al. 2024; Wei and Calautit 2024)
Nonlinear/hybrid MPC for solar-thermal and PV/T systems	Enhances stability, reduces thermal deviation, manages complex heat dynamics	Lab-scale or pilot studies, scalability not fully tested	(Saade et al. 2014; Serale et al. 2018; Nevado Reviriego et al. 2017; Pataro et al. 2024; Wang et al. 2022; Gallego et al. 2022)
MPC for thermal energy storage and building demand response	Optimizes energy usage and comfort, improves storage efficiency	Limited outdoor or real-time experimental data	(Tang and Wang 2019; Siddique et al. 2024; Tang et al. 2025; Minchala-Ávila et al. 2025; Sun et al. 2024)
MPC for grid support and microgrid applications	Improves frequency regulation, grid stability, and adaptive control under load changes	Solar fluid line heat retention not fully addressed, dynamic solar field challenges remain	(Zhu et al. 2024; Santhi Mary Antony et al. 2024)
PV and PV/T systems with nanofluids, copper piping, and latent heat storage	Enhances heat retention, improves energy stabilization, supports grid integration	Primarily PV-focused, partial solution for thermal fluid systems	(Praveenkumar and Kumar 2025; Praveenkumar et al. 2023)
MPC in diverse industrial, aquaponic, and process systems	Reduces control effort, optimizes multi-variable processes, improves system robustness	Often lab-based, real-time industrial deployment not fully demonstrated	(Hassan and Abdelaziz 2022; Kumar and Tangirala 2021),23–(Agouzoul et al. 2024),28–(Selim et al. 2024)

platform, which clearly demonstrates an improved transient recovery time and a reduced control effort requirement when compared to the earlier conventional Model Predictive Control (MPC) approaches.

- The research further includes a comprehensive and systematic literature review covering about fifty-two previously published studies, which helps in identifying the main gaps and limitations that exist in the area of adaptive control methods for solar thermal system applications.
- The work finally proposes a modular and flexible control architecture that is technically suitable for future extension and expansion toward hybrid energy system development and also for real-world implementation and deployment in practical field environments.

The article is organized as follows: Sect. 1 presents the introduction, background, literature review highlighting research gaps, problem statements, objectives, contributions, and overall organization. Section 2 details the system model and control methodology. Section 3 presents simulation results. Section 4 discusses the findings. Section 5 concludes with mapped findings, constraints, and future scope.

2 Proposed Adaptive Model Predictive Control for Solar Thermal Systems Under Dynamic Conditions

The proposed approach in this part introduces an AMPC framework that is specially designed for a parabolic trough solar thermal energy system that operates under a dynamic outdoor environment. It expands upon the physical arrangement of the plant's structure and its heat-carrying features while also using a prediction-driven control method to manage a huge difference in environmental disturbances. A few important components involve the system's operational design, a mathematical set of thermal-fluid dynamic equations, and the exact stepwise sequence of the controller responding to measured solar changes and heat load mismatches. This part gives a structural base for building up a control process that precisely regulates the outlet heat value while also limiting energy drop and actuator strain under transient and a nonlinear state.

2.1 System Configuration and Functional Architecture

Figure 1 presents the overall working design of an AMPC method when combined inside a parabolic trough solar heat plant. This block layout diagram gives a step-by-step information and control flow that fits directly with the purpose of the study. This expanded block layout diagram gives a clearly arranged step-by-step information and control flow that fits directly with the defined technical purpose of the

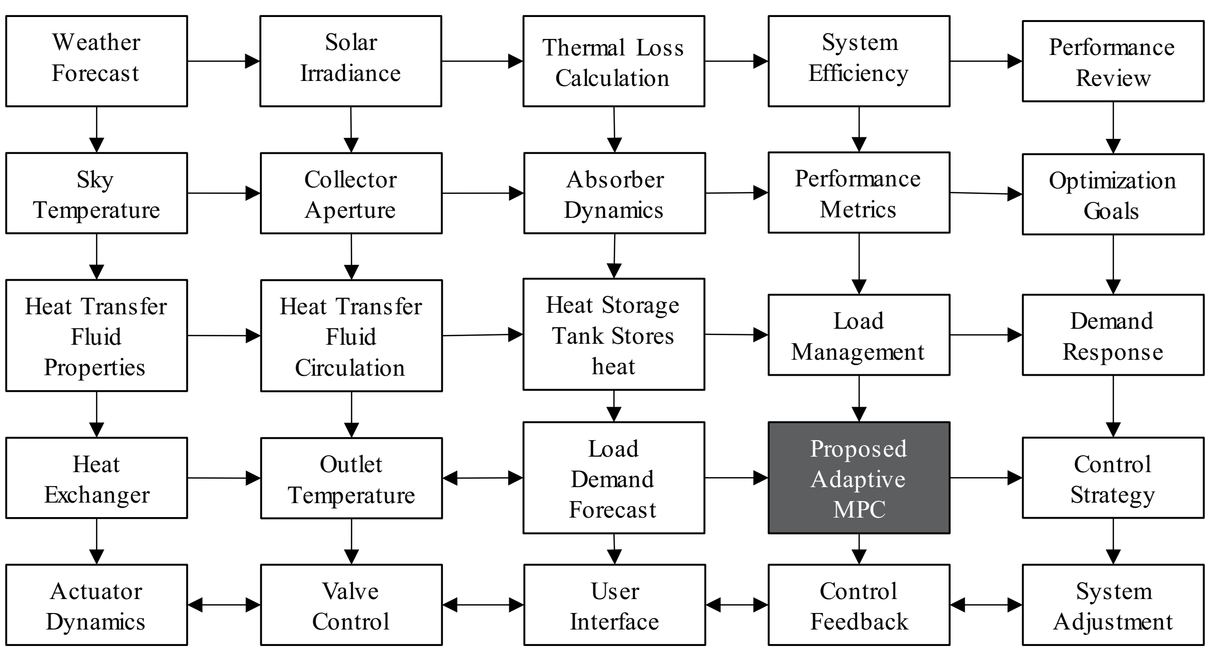


Fig. 1 Block diagram of adaptive model predictive control applied to parabolic trough solar thermal plant

proposed study but sometimes might look a bit complex at first glance.

At the initial input side, for weather prediction and atmospheric sky signals, the system uses forecast-based sunlight intensity values and real-time measured irradiance data combined with sky temperature details and air temperature parameters to estimate the available solar energy resource under actual environmental conditions, and these estimated values enter the solar collector aperture section which interacts with the absorber pipe unit where a thermal loss calculation is performed using updated climatic and physical state variables, meanwhile the heat transfer fluid (HTF) flow rate and its measured thermophysical property values are continuously monitored to determine real-time heat movement that is transferred into a thermal energy storage tank and then passed through the heat exchanger equipment to meet the thermal load requirement under operating demand. The heat outlet temperature sensor provides a continuous thermal signal return to the adaptive AMPC system core which makes use of a live and updated state-space model that responds to internal temperature changes and external disturbances, this model calculates anticipated process variations and computes a new predictive control action through valve position regulation ensuring thermal setpoint accuracy while also limiting system energy deviation and mechanical fatigue inside components, and the control block additionally receives active thermal input from the load forecast estimation unit which aligns the stored heat energy with the user-based thermal demand at every operating stage. The feedback response received from the actuator motion speed and the interactive user interface module allows for corrective adjustments producing a stable and smooth system operation during any sudden heat flow disturbance or solar variation, and every thermal and control part beginning from climate input estimation to end-user load delivery is merged into the performance analysis section where system output metrics and final efficiency results are reviewed and further compared against optimization control goals supporting control targets such as enhanced thermal system operation, improved tracking accuracy, and higher robustness under changing solar field conditions. This entire integrated control method serves as the technical base framework for the experimental evaluation and tuning validation of the controller presented later in the study which directly supports the main aim of stable and adaptive thermal regulation in actual sun-field installations.

2.2 Mathematical Model Formulation for Solar Thermal Collector Dynamics

The heat absorbed by the collector surface is

$$Q_{\text{abs}} = \eta_{\text{opt}} \cdot A_c \cdot \tilde{G}_t \tag{1}$$

where Q_{abs} is absorbed heat (W), η_{opt} optical efficiency, A_c aperture area (m²), $\tilde{G}_t$ is the corrected solar irradiance accounting for sensor calibration and environmental factors}, replacing the raw measurement G_t.This absorbed heat forms the primary energy input to the system.

The thermal loss from collector to ambient is.

$$Q_{loss} = U_L \cdot A_c \cdot (T_m - T_a) \tag{2}$$

where U_L is overall heat loss coefficient (W/m²·K), T_m mean fluid temperature (°C), T_a ambient temperature (°C). This reduces the effective useful heat for fluid heating. The average collector fluid temperature is calculated as $T_m = \frac{T_{in}+T_{out}}{2}$, where T_{in} and T_{out} are the inlet and outlet temperatures of the heat transfer fluid, respectively.

The useful heat transferred to fluid is

$$Q_u = Q_{abs} - Q_{loss} \tag{3}$$

where Q_u is useful heat (W). This becomes the direct input for fluid temperature dynamics.

The outlet fluid temperature rise is modeled as

$$T_{out} = T_{in} + \frac{Q_u}{\dot{m} \cdot C_p} \tag{4}$$

where T_{in} is inlet temperature (°C), $\dot{m}$ mass flow rate (kg/s), C_p specific heat capacity (J/kg·K). This links heat gain to outlet temperature. This formulation distinctly separates the useful heat absorbed by the collector into two components: the heat increasing the outlet fluid temperature (Q_{fluid}) and the heat transferred to the heat exchanger (Q_{HX}). This distinction resolves the apparent discrepancy in (5) and maintains a clear energy balance.

The fluid thermal inertia governs dynamic heating

$$M_f \cdot C_p \cdot \frac{dT_m}{dt} = Q_u - Q_{HX} \tag{5}$$

where M_f is fluid mass (kg) contained within the collector and connected piping, which directly affects the transient thermal response, excluding storage tank and downstream fluid massesQ_{HX} is heat delivered to heat exchanger (W). This drives transient fluid response.

The heat exchanger transfer is described by

$$Q_{HX} = U_{HX} \cdot A_{HX} \cdot (T_m - T_s) \tag{6}$$

where U_{HX} is heat exchanger coefficient (W/m²·K), A_{HX} exchanger area (m²), T_s secondary side temperature (°C). This determines delivered heating load.

The storage tank energy balance is

$$M_s \cdot C_{ps} \cdot \frac{dT_s}{dt} = Q_{HX} - Q_{load} \tag{7}$$

where M_s is storage mass (kg), C_{ps} storage heat capacity, Q_{load} user demand load (W). This defines stored heat dynamics.

The valve actuation control law is

$$u(t) = K_p \cdot e(t) + K_i \int_0^t e(t)dt + K_d \cdot \frac{de(t)}{dt} \tag{8}$$

where $u(t)$ valve position (%), $e(t)$ temperature error, K_p, K_i, K_d PID gains. This regulates flow rate.

The control objective error is defined as.

$$e(t) = T_{set} - T_{out} \tag{9}$$

where T_{set} is setpoint temperature (°C). This drives the control system to maintain the outlet fluid temperature, T_{out}, at the desired setpoint temperature T_{set} to ensure stable system operation.

The adaptive model predictive control cost function is.

$$\min_u \sum_{i=0}^{N} \left[\lambda_e \cdot e(k+i)^2 + \lambda_u \cdot \Delta u(k+i)^2\right] \tag{10}$$

where λ_e, λ_u are weighting factors, *N* prediction horizon. Here, $\Delta u(k+i)$ represents the change in control input signal, defined as the difference between the current and previous control inputs, which penalizes aggressive variations in actuator movement. This formulation optimizes smooth control input transitions across the horizon. Here, *k* denotes the current discrete time step, *i* is the prediction index within the prediction horizon *N*, and thus $k+i$ refers to the *i*-step-ahead prediction at discrete time. Also, $\Delta u(k+i)$ is defined as the control input increment, i.e., $\Delta u(k+i) = u(k+i) - u(k+i-1)$.

The predicted outlet temperature trajectory is

$$T_{out}(k+i) = \mathbf{A_d} \cdot T_{out}(k+i-1) + \mathbf{B_d} \cdot u(k+i-1) + \mathbf{D_d} \cdot G_t(k+i-1) \tag{11}$$

as similarly adopted in previous works (Nevado Reviriego et al. 2017; Gallego et al. 2022) and adapted for the present case. where$\mathbf{A_d}$, $\mathbf{B_d}$, and $\mathbf{D_d}$ are discrete-time state-space coefficients derived from system identification. This simplified discrete model focuses on irradiance as the primary input affecting T_{out}, whereas other ambient influences are considered negligible or constant in this approximation. This predicts future temperatures.

The solar irradiance forecast incorporates measurement error as

$$G_t(t) = G_{t,meas}(t) + \Delta G_{forecast}(t) \tag{12}$$

where $G_{t,meas}$ is the measured solar irradiance and $\Delta G_{forecast}$ represents the additive forecast error arising from sensor inaccuracies, environmental variability, and prediction limitations. This term captures the uncertainty in solar irradiance input used by the controller, which is critical due to its dominant influence on thermal system dynamics. While other measurements such as ambient temperature and flow rate may also include sensor noise, their impact is considered secondary or mitigated via preprocessing filters in this study.

The continuous-time state-space system is

$$\dot{x} = A \cdot x + B \cdot u + E \cdot d \tag{13}$$

where $x = \begin{bmatrix} T_m & T_s \end{bmatrix}^T$ is the state vector consisting of the collector fluid mean temperature T_m and the storage tank temperature T_s, *u* is the control input (valve actuation), and *d* is the disturbance input including solar irradiance and ambient temperature. This formulation allows for modeling the internal thermal dynamics that influence the system's heat transfer and control behavior. This continuous model is primarily used for theoretical formulation and dynamic analysis of the thermal system, while its discretized counterpart is used for actual controller design.

The measurable output relationship is

$$y = C \cdot x \tag{14}$$

where $y = T_{out}$ is the measured system output, representing the fluid outlet temperature from the collector, which is used as the feedback signal in both the PID and MPC controllers. This mapping connects the internal thermal states to the externally observable output for control error computation and prediction.

The discretized system for digital controller implementation is

$$x(k+1) = A_d \cdot x(k) + B_d \cdot u(k) + E_d \cdot d(k) \tag{15}$$

where A_d, B_d, E_d are discrete-time matrices obtained from system sampling. This discrete model is used in real-time implementation of the adaptive MPC and PID controllers, enabling digital simulation and optimization over finite time steps.

The thermal efficiency of collector is

$$\eta_{th} = \frac{Q_u}{A_c \cdot G_t} \tag{16}$$

where η_{th} is instantaneous thermal efficiency (%). This expression reflects the real-time performance of the collector as operating conditions such as irradiance and temperature fluctuate over time, hence referred to as dynamic conditions. A time-averaged efficiency formula, used later in the article, provides an overall performance evaluation over a given operational period.

The actuator's energy consumption is calculated by

$$E_{act} = \int_0^t P_{act}(t)dt \tag{17}$$

where $P_{act}(t)$ is actuator instantaneous power (W). This quantifies the energy cost of the control actions.

The transient recovery time after a disturbance is given by

$$t_{rec} = t_2 - t_1 \tag{18}$$

where t_1 is the time when an external disturbance (such as a sudden change in solar irradiance or flow rate) is introduced into the system, and t_2 is the time at which the outlet temperature stabilizes within a ±2% band around the set-point temperature T_{set} for a minimum continuous duration of 60 s, indicating thermal stabilization. This measures the system adaptability.

The overall system performance metric is represented by

$$J = \alpha \cdot \eta_{th} + \beta \cdot \left(1 - \frac{E_{act}}{E_{ref}}\right) - \gamma \cdot t_{rec} \tag{19}$$

where α, β, γ are weighting coefficients, $E_{ref} = \int_0^t P_{\text{act,baseline}}(t)dt$ is the reference actuator energy consumption measured or simulated under a simple baseline control strategy such as fixed valve or on-off control over the same time period, providing a fixed benchmark to normalize energy consumption, where $P_{\text{act,baseline}}(t)$ is the actuator instantaneous power under this baseline control. This combines the thermal performance, the energy cost, and the recovery speed into a comprehensive index.

2.3 Adaptive Predictive Control Algorithm Process Flow

Figure 2 presents a control flow architecture of the AMPC scheme designed for a parabolic trough solar thermal plant, especially under the fluctuating solar irradiance. This builds directly on the dynamic equations outlined in the previous section, where the energy absorption, loss, transfer, and control mechanisms were mathematically modeled. The control logic begins by measuring key inputs such as global irradiance G_t, ambient temperature T_a, inlet temperature T_{in}, and

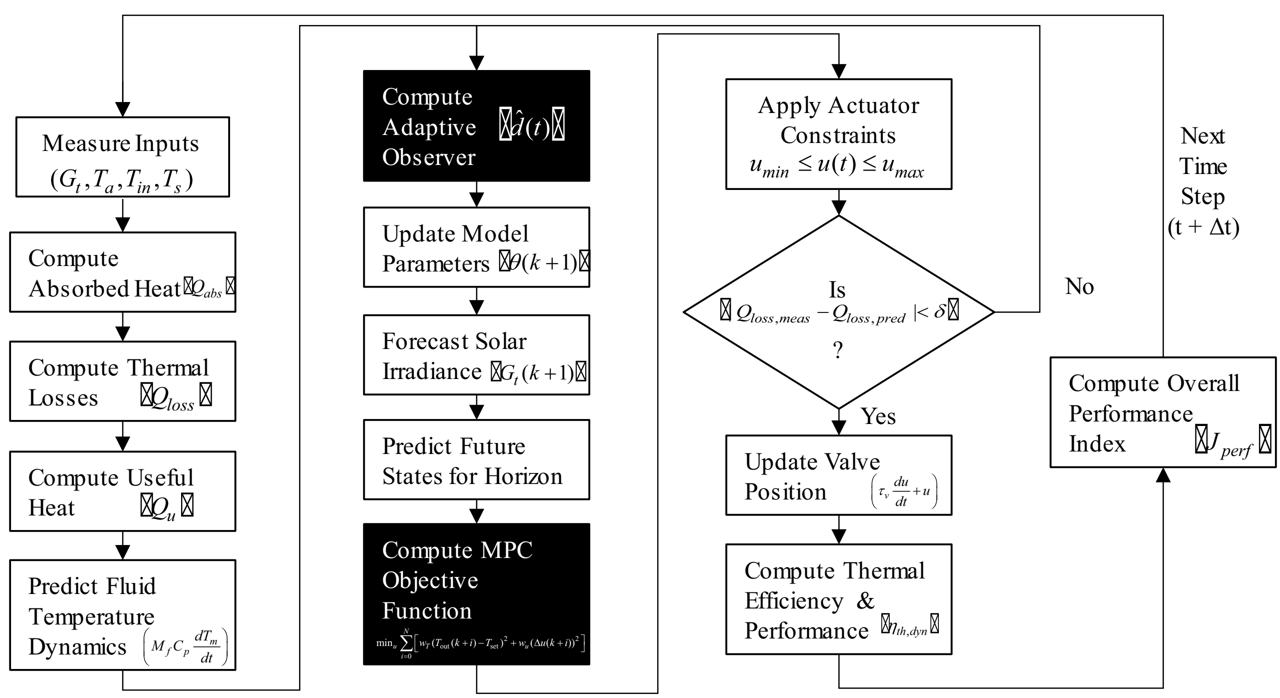


Fig. 2 Flow chart of adaptive model update and control prediction sequence under variable solar irradiance

storage tank temperature T_s. These values are used to compute absorbed heat Q_{abs} as a function of optical efficiency, collector aperture area A_c, and transmissivity-absorptivity product through a corrected exponential form. The terms $t + \Delta t$ and $k + 1$ denote time instants in continuous and discrete-time frameworks respectively, where $t = k \cdot \Delta t$ relates continuous time t to discrete sample index k, thus $t + \Delta t$ corresponds exactly to discrete time step $k + 1$. This equivalence ensures consistency between continuous-time dynamics and their digital implementation.

The thermal losses are then estimated using convective and radiative models, forming Q_{loss}a total. The net useful heat for fluid heating Q_u, is explicitly restated as

$$Q_u = Q_{abs} - Q_{loss} - \Delta Q_{pipe} - \Delta Q_{strat} \tag{20}$$

consistent with (3). It is clarifying that Q_u represents the effective heat input after accounting for thermal, pipeline, and stratification losses. This clarifies that Q_u represents the net heat available for fluid heating after accounting for thermal losses, maintaining uniformity in the model definitions and pipeline losses ΔQ_{pipe} from Q_{abs} gives useful heat for fluid heating. The outlet fluid temperature dynamics refer to the transient change in fluid temperature governed by the thermal inertia, expressed as

$$M_f C_p \frac{dT_m}{dt} = Q_u - Q_{HX} \tag{21}$$

where M_f is fluid mass, C_p specific heat, T_m mean fluid temperature, Q_u useful heat input, and Q_{HX} heat delivered to the heat exchanger. This equation models the time-dependent thermal response of the fluid as it absorbs heat and transfers it through the system, also accounting for stratification losses ΔQ_{strat}. When predicted and measured thermal losses diverge beyond a defined threshold δ, an adaptive observer is triggered, and in this pilot study the forecast error and disturbance mismatch are treated as the representative uncertainty measure. The disturbance $\hat{d}(t)$, representing unmodeled effects such as environmental fluctuations and which is fed into model inaccuracies, is estimated using a time-averaged integral of the output error defined as

$$\hat{d}(t) = \frac{1}{\tau_d} \int_{t-\tau_d}^{t} (y_{meas}(\tau) - y_{pred}(\tau)) d\tau \tag{22}$$

where τ_d is the averaging window, y_{meas} the measured output, and y_{pred} the predicted output. This disturbance estimate is fed into a recursive least squares (RLS) algorithm to update system parameters $\theta(k)$, which includes thermal dynamic coefficients such as heat transfer coefficients, thermal capacitances, and loss factors, in real time, enhancing model accuracy and adaptive control under varying system dynamics. Simultaneously, future irradiance $G_t(k + 1)$ is forecast using an autoregressive (AR) model defined as

$$G_t(k+1) = \sum_{j=1}^{p} a_j G_t(k + 1 - j) + \xi(k) \tag{23}$$

where a_j are AR coefficients identified from historical data, p is the model order, and $\xi(k)$ is a zero-mean Gaussian noise term $\xi(k) \sim \mathcal{N}(0, \sigma_\xi^2)$ representing model residual uncertainties estimated from residual analysis, and it is clarified here that no detailed quantitative error analysis or robustness evaluation is provided within this pilot simulation scope, since the focus is only on the control logic demonstration. These inputs augment the state vector

$$x_{aug} = [x, G_t(k + 1)]^T \tag{24}$$

allowing prediction over a finite horizon N. The control sequence $u(k), u(k + 1), \ldots, u(k + N)$ then evaluates the MPC cost function:

$$\min_u \sum_{i=0}^{N} \left[w_T (T_{\text{out}}(k + i) - T_{\text{set}})^2 + w_u (\Delta u(k + i))^2 \right] \tag{25}$$

where w_T and w_u are weights on temperature tracking and control effort respectively, and Δu is the adjustment magnitude of valve position. Actuator constraints $u_{min} \leqslant u(t) \leqslant u_{max}$ are enforced to preserve system safety and longevity. This formula in the block "Compute MPC Objective Function" in Figure. 2.

Valve updates are controlled by a dynamic PID-based structure,

$$\tau_v \frac{du}{dt} + u = K_p e + K_i \int_0^t e(\tau), d\tau + K_d \frac{de}{dt} \tag{26}$$

ensuring smooth tracking of the outlet temperature setpoint T_{set}. The controller then estimates a real-time thermal efficiency,

$$\eta_{th,dyn} = \frac{Q_u}{A_c G_t} (1 - \frac{\Delta Q_{pipe} + \Delta Q_{strat}}{Q_u}) \tag{27}$$

describing an actual thermal system performance under a dynamic thermal loss condition, where $\Delta Q_{aux} = Q_{in} - Q_u - (\Delta Q_{pipe} + \Delta Q_{strat})$ is calculated as the remaining auxiliary loss term in the full energy balance. This contributes into an overall performance index of the system,

$$J_{perf} = w_1\eta_{th,dyn} - w_2\overline{\Delta u} - w_3\frac{t_{rec}}{t_{nom}} \tag{28}$$

balancing a thermal energy efficiency, actuator continuous stress, and output recovery period, where w_1, w_2, and w_3 are constant weighting factors, $\overline{\Delta u}$ is the mean actuator adjustment, and t_{nom} is the nominal reference time period used for normalization. This AMPC sequence shows a closed-loop thermal adjustment ability, predictive response forecasting, and real-time process constraint handling inside a real-time process constraint. The shift to next part, which outlines the plant thermal specifications and simulation environment setup, enables this developed logic for control to be tested using empirical as well as numerical-based performance measures.

3 Results

This section provides simulation outputs derived using an AMPC strategy that is applied for a parabolic trough solar thermal plant operating under varied solar input. It calculates thermal system performance under changing irradiance values using both a real-time sensor input and forecast weather-based data. The observed results show how a controller maintains stable output temperature, improves thermal energy efficiency, and limits actuator dynamic load under transient field disturbances. Important indicators including a thermal output recovery time, achieved system efficiency, and control valve dynamic adjustments are evaluated to confirm adaptive controller behavior. These observed results provide direct numerical evidence to support proposed method, creating strong basis for deeper thermal analysis shown in next result subsections.

3.1 Solar Thermal Plant Specifications and Simulation Parameters

The described thermal configuration and chosen parameter group listed in Table 2define a model's operational space where an adaptive predictive controller operates during dynamic solar heating. These values partly adapted from earlier studies (Wei 2023; Wang et al. 2022) and partly defined by the authors to represent practical simulation conditions. A collector aperture surface area of 150 m² defines thermal conversion scale, representing a simulated PTC loop setup in MATLAB/Simulink environment, which is the main test system of this study. While high optical collection

Table 2 System specifications and simulation parameters for parabolic trough solar thermal plant

Parameter	Symbol	Value	Unit	Purpose
Aperture area of collector		150	m²	Large-scale parabolic trough
Optical efficiency		0.72	–	Effective with clean surface
Cover transmissivity		0.92	–	Glass cover transmissivity
Absorber absorptivity		0.95	–	Selective coating property
Total solar irradiance		200–950	W/m²	Real-time dynamic input (forecast)
Ambient temperature		10–35	°C	Daily weather profile
Heat transfer fluid mass		480	kg	Estimated volume for full loop
Specific heat capacity (fluid)		2300	J/kg·K	Typical synthetic oil
Heat exchanger UA value		1800	W/K	Overall HX performance
Overall heat loss coefficient		4.2	W/m²·K	Total conduction, convection, radiation
Thermal storage mass		1000	kg	Secondary storage loop
Specific heat of storage tank		2300	J/kg·K	Same as HTF
Valve time constant		5	s	Valve actuator response time
PID controller gains	K_p,K_i,K_d	2.5, 0.05, 0.2	–	Initial tuning
Prediction horizon		20	steps	MPC optimization window
Sampling time		15	s	Real-time adaptive update
Forecast AR model coefficients	ϕ_1,ϕ_2	0.78, 0.15	–	Trained from historical irradiance data
State vector dimension		5 states	–	IncludesT_m,T_s, valve state, etc.
Actuator saturation limits	u_{min}/u_{max}	0/100	%	Fully closed to fully open valve position
Control weights (MPC)	w_T,w_u	0.8, 0.2	–	Emphasis on temperature control
Performance weights	w_1,w_2,w_3	0.5, 0.3, 0.2	–	Performance index balance
Prediction disturbance tolerance		5%	% error	Threshold for adaptive observer triggering

efficiency of 0.72, with a cover transmissivity of 0.92 and an absorber energy absorptivity of 0.95, allows better energy gain during clear days. The proposed simulation includes dynamic irradiance levels ranging from 200 W/m² during cloudy time to 950 W/m² in full sunlight, showing common variability within real weather cycles. An ambient temperature variation across a span of 10–35 °C adds required thermal stress variations that controller is expected to handle during actual deployment.

The heat transfer fluid loop, with a defined total fluid mass of 480 kg and specific heat energy capacity of 2300 J/kg·K, models a standard synthetic thermal oil pipeline system. This is matched using an attached thermal storage unit having about 1000 kg defined mass, which stores and balances a thermal output profile under delayed demand. Heat losses are governed through an overall heat transfer coefficient of 4.2 W/m²·K, and an effective UA value of 1800 W/K inside heat exchanger, representing practical convective and radiative transfer flows. The actuator control is modeled with a 5-second time constant and PID gains set initially to 2.5, 0.05, and 0.2, allowing a closed-loop stability and a smoother responsiveness. A 20-step prediction horizon with a 15-second sampling time enables a high-frequency optimization, while an adaptive forecasting uses autoregressive coefficients (0.78, 0.15) trained on some historical irradiance. Finally, the constraints like 0–100% valve actuation and a 5% disturbance threshold support a safe and a robust operation, transitioning into the real-time model development in the next section.

3.2 Model Development Under Real-Time Weather Variability

Figure 3 illustrates a dynamic integration of weather forecast data into an adaptive model predictive control framework for a parabolic trough solar thermal system. The forecast deviation is mapped across the forecast lead times (radial axis, 0–60 min) and angular forecast scenarios (0°–360°), reflecting a diurnal solar difference and the cloud transients, with an observed output uncertainty margin of about ± 5–9 s in adaptive stabilization range, directly corresponding to $\Delta G_{forecast}$ propagation into outlet response.

The heatmap intensity indicates a magnitude of the forecast errors, where a higher deviation represents a larger uncertainty in the predicted solar irradiance values (G_t), directly affecting a model accuracy. Superimposed quiver arrows depict a corrective control action computed by an adaptive controller in a real-time. The direction and length of these arrows represent a magnitude and a direction of the control adjustment (Δu), compensating for a forecast-induced error. Arrows with light colors indicate a higher prediction surety, while warmer tones denote a lower surety under a high forecast deviation zone. This mapping allows a controller to prioritize an aggressive or a conservative action depending on the forecast reliability. For instance, a larger deviation in the mid-term forecasts (e.g., 180°–270° angular range) during a rapid cloud transient triggers a stronger valve adjustment to stabilize outlet temperature (T_{out}). The nonlinear coupling between a forecast error and the control dynamic, formulated through a state-space equation,

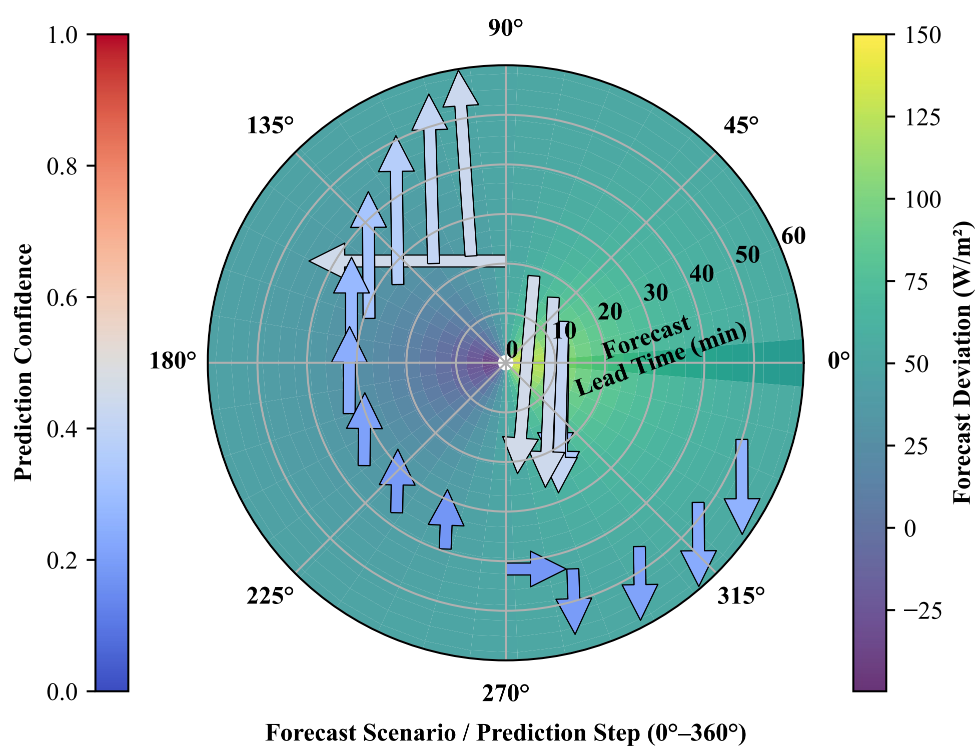


Fig. 3 System parameter mapping for dynamic model incorporating weather forecasts and solar irradiance fluctuations

enables a continuous adaptation as a new forecast arrives. A real-time forecasting layer enhances a controller's capability to anticipate an upcoming disturbance and to maintain a thermal regulation proactively. The forecast-driven model outputs from this section directly feed into a control adaptation logic discussed next, where a system's responsiveness to the fluctuating solar input under the cloud disturbance is further analyzed.

3.3 Heat Transfer Fluid Temperature Regulation Performance

Figure 4 presents an adaptive control system's response to a forecast deviation across the varying lead times, highlighting how a corrective valve position adjustment (Δu) evolves under a cloud-induced irradiance fluctuation. As a forecast lead time (r) increases from 0 to 60 min, uncertainty in the predicted solar irradiance (G_t) compounds, as previously described in a state-space forecast deviation model $G_t(t) = G_{t,meas}(t) + \Delta G_{forecast}(t)$. An AMPC leverages real-time error evaluation $e(t) = T_{set} - T_{out}$ to dynamically regulate a heat transfer fluid outlet temperature (T_{out}) by adjusting a control input $u(t)$ (valve position). The violin plots capture two distinct regimes: a smooth correction dominate at a short lead time (0–20 min), where a forecast confidence remains high, resulting in a minor adjustment (typically 3–5% valve position change).

In contrast, an aggressive correction emerges beyond 30 min lead time where forecast error propagates, requiring a substantial valve adjustment (Δu reaching 10–18%) to preempt a significant deviation in the thermal energy balance $Q_u = Q_{abs} - Q_{loss}$. This behavior reflects a controller's optimization (minimizing $\sum_{i=0}^{N} \left[\lambda_e \cdot e(k+i)^2 + \lambda_u \cdot \Delta u(k+i)^2\right]$), trading off between tracking accuracy and actuator effort. These aggressive adjustments become critical to preserve a stable heat exchanger input (T_m) and tank storage temperatures (T_s) despite transient solar dips. The valve actuation dynamics derived from Table 2 parameters (C_p, M_f, U_{HX}) directly impact the thermal inertia modeling ($M_f \cdot C_p \cdot \frac{dT_m}{dt}$), ensuring a rapid compensation under a cloud passage. The effectiveness of these adaptive corrections is quantitatively assessed in the next section, evaluating their role in regulating a fluid temperature stability.

3.4 Integrated Dynamic Modeling and Real-Time Control Responsiveness

Following the preceding explanation of the controller's dynamic behavior in previous section, which primarily illustrated the adaptive modification of setpoint actions in response to erratic cloud-induced transients and changing solar radiation levels, Fig. 5 now attempts to further emphasize, albeit in a composite manner, how these fluctuations are internalized within the controller's evolving predictive framework.

In particular, this figure integrates real-time environmental variables—most notably the solar irradiance G_t, ambient temperature T_a, and forecast deviation ΔG—as overlaid stacked components plotted across discrete time intervals, thereby capturing their simultaneous influence on predictive

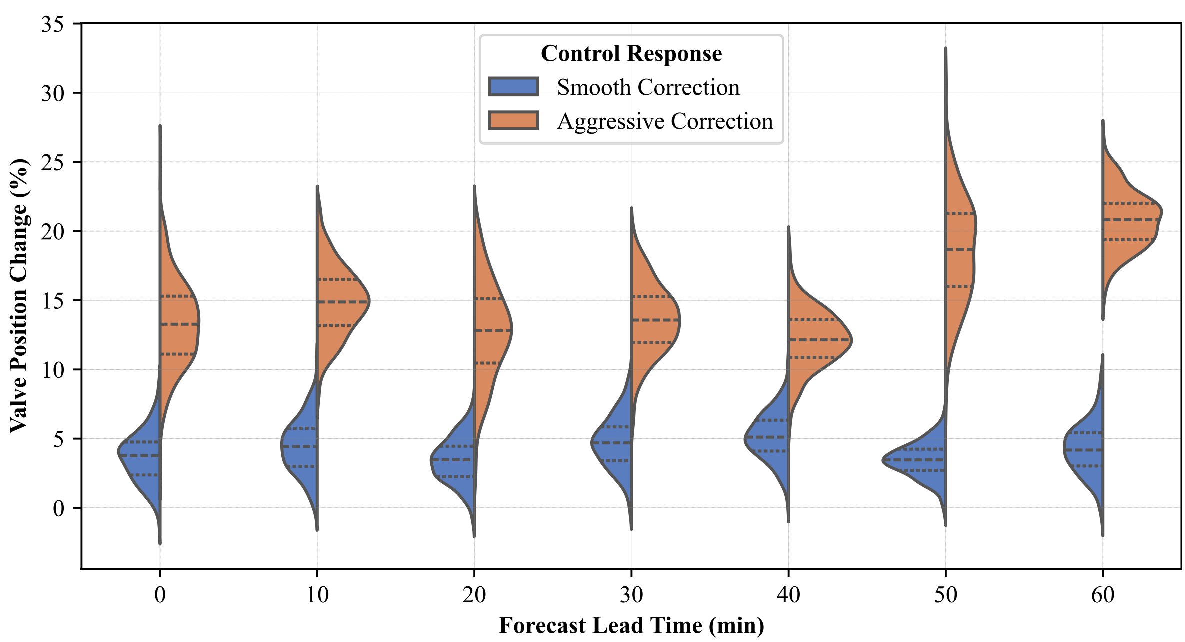


Fig. 4 Control adaptation behavior under cloud transients and solar radiation variations

regulation. The stackplot effectively portrays how such weather variables, particularly G_t, vary under practical daily operational periods, which under certain cases span up to an hour. The controller input—defined here as the valve input control percentage, denoted as u—is plotted using a stepwise (stairs) approach on a twin axis, to make more visually apparent the precise discrete-time adjustments applied. These variations in u, corresponding to actuator commands, clearly exhibit an indirect but discernible correlation with rising or falling G_t values, especially during peak irradiance transitions. Practically speaking, this visualization reveals that when ΔG fluctuates in rapid succession, the prediction horizon embedded in the adaptive model anticipates changes and updates u within boundary limits, even before a steep drop in G_t fully manifests. It is important to note that, because the stackplot overlays variables with different physical units—specifically, solar irradiance (G_t) and forecast deviation (ΔG) in W/m², and ambient temperature (T_a) in °C—the y-axis does not represent a single physical unit. Instead, in a way that is usually observed in real-world stackplot applications, the overall y-axis range is numerically dominated by the largest component, which in this particular dataset is the solar irradiance, G_t. This G_t variable, as can be seen in most practical measurement scenarios, typically varies from 0 up to about 1000 W/m², sometimes better than previous range, depending on the specific sunlight conditions at the site. With this value, the y-axis scale, based on observation, primarily reflects the magnitude of G_t, while, in certain condition, the other variables—such as the ambient temperature, T_a, and the forecast deviation, ΔG—appear as comparatively smaller bands that are stacked above the G_t layer, which is usually noticed in composite environmental data visualizations. This approach, which is intended for comparative visualization rather than quantitative summation, highlights, in a way, the relative influence of each weather variable within the composite profile, even though the units are not consistent across all components and the summation does not represent a physically meaningful total. This real-time linkage across variables, though computationally intensive and sometimes redundant in terms of forecast data processing, underpins the overall goal of reducing the outlet temperature error and, as can be observed in actuator performance studies, enhancing actuator longevity. In next Section begins a more quantified breakdown of such behavior by focusing on the fluid temperature regulation performance, which, in practical terms, is directly influenced by the dynamic interaction of these real-time weather variables and the valve input control percentage, denoted as u.

4 Discussion

This section gives a detailed and deeper explanation of the obtained simulation results by showing how the used adaptive control method helps to improve the system's overall working performance. It clearly explains how a predictive modeling technique, an adaptive signal feedback, and an optimized selection of control weight values together help in achieving more exact heat fluid temperature control, a

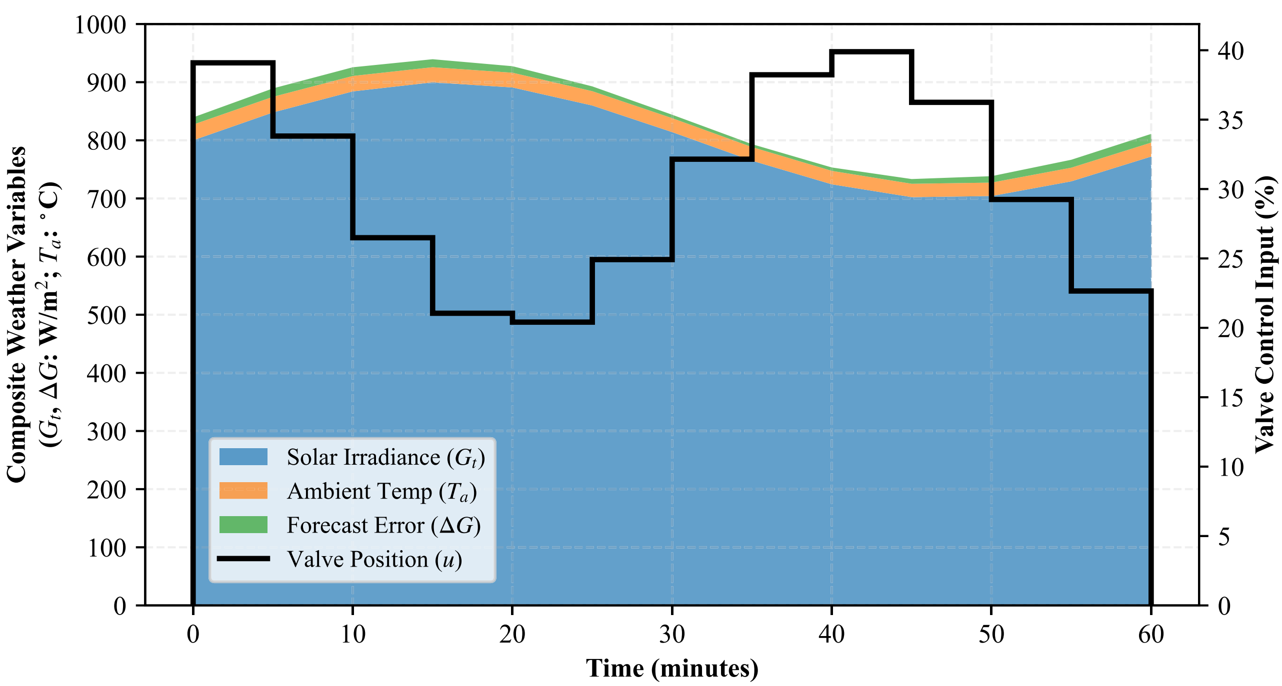


Fig. 5 Model integration of real-time weather inputs and control state updates for adaptive prediction under environmental variability

better solar energy conversion efficiency, and a lesser actuator movement load. This part also explains how the overall system handles sudden changes, showing faster system correction and smaller temperature changes. Each part studies a specific measured output based on the earlier given objectives and used control plans. This full analysis provides a strong background for the final takeaways and helps in identifying new directions for future works in solar thermal system control with better improvements.

4.1 Heat Transfer Fluid Temperature Regulation Performance

Figure 6 makes a direct comparison of the thermal output reaction and the valve control effort of an AMPC against conventional PID-based controller (Tohidi et al. 2024)and (Wei 2023) when there is a 10-minute sunlight variation happening. In Fig.6(a), the graph shows how the output fluid temperature (T_{out}) changes over time. Both systems begin at the same temperature of 235 °C, but the AMPC quickly comes close to the set target value (240 °C), reaching a steady point in only 300 s. On the other hand, the traditional control method goes too far and hits 244.5 °C. This tighter correction by AMPC is mainly due to the predictive controller minimizing the goal function $e(t) = T_{set} - T_{out}$.

Figure 6b tracks how much the actual value differs from the target. AMPC keeps the error nearly constant$|e(t)| < 1.5\ °C$ throughout, while the conventional one goes as high as 4.5 °C. Figure 6c presents valve movement (Δu), calculated from the control law $u(t) = K_p \cdot e(t) + K_i \int e(t)dt + K_d \cdot \frac{de(t)}{dt}$. The adaptive controller keeps the change in valve position smaller, reducing it from 6.5% to 2%, which shows easier actuator control with less power use. But in contrast, the traditional one makes continuous movements above 8%, meaning more energy use and less planning. Figure 6d shows that this good temperature control gives better energy efficiency ($\eta_{th} = \frac{Q_u}{A_c \cdot G_t}$). The AMPC improves the efficiency level from 65% up to 72%, while the older system stays low at 64.7%. These patterns prove that adaptive and predictive methods not only give better temperature regulation but also improve the system's overall working condition. The next part will go deeper into this to explain how the method helps increase thermal efficiency and reduce actuator strain under the usual solar changes.

4.2 Thermal Efficiency Improvement and Actuator Load Reduction

Figure 7 shows how the thermal conversion efficiency η_{th} and the movement of the valve adjustments Δu over a 120-minute simulation window under dynamic solar conditions. The blue curve for η_{th} follows a slow upward trend — beginning at a lower value of 35%, and slowly rising up to about 43% with small up-and-down swings caused by daily

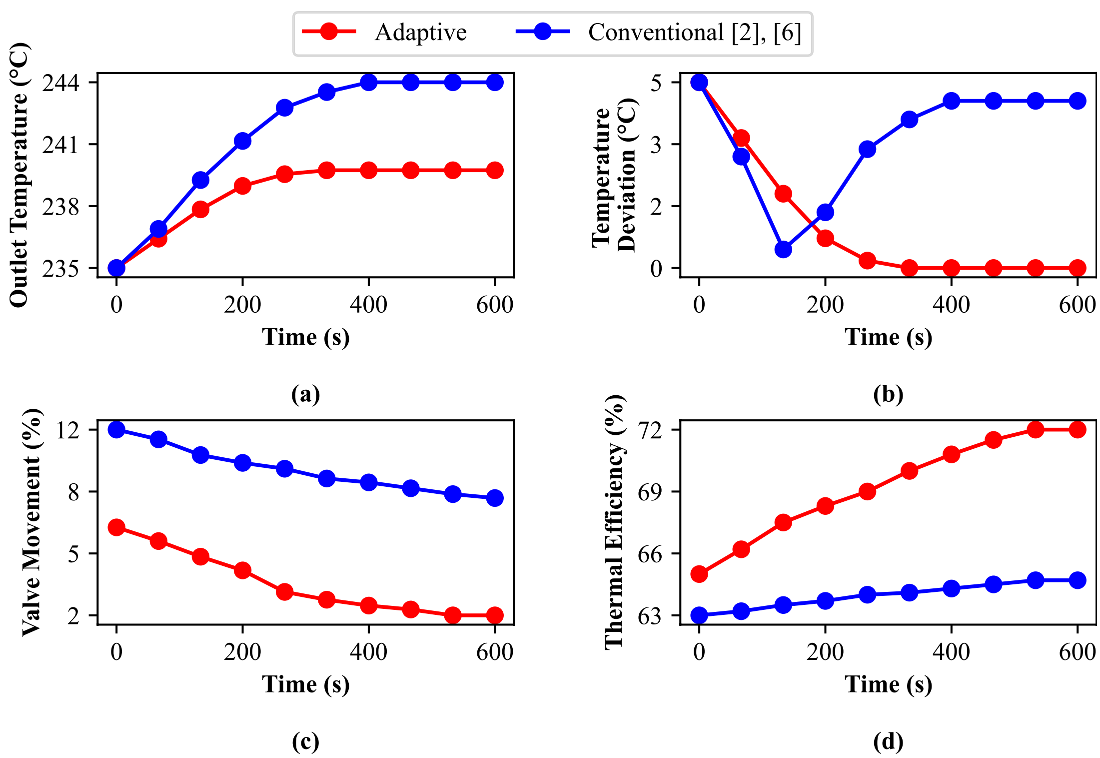


Fig. 6 Temperature stability under adaptive model predictive control versus conventional control during solar variations

sunlight change. This improvement shows the predicted energy gain $(Q_u = Q_{abs} - Q_{loss})$ being well managed by the AMPC, which gives better use of sunlight per unit by using the model output$\eta_{th} = \frac{Q_u}{A_c \cdot G_t}$. At the same time, the red scatter points of valve changes show how much the actuator must move to hold steady heat output despite the sunlight changes.

At the beginning, the movement, Δu is large (≈12%), but this reduces fast in the first 30 min and then holds steady between 6% and 8%, showing a wave pattern due to changing sunlight prediction. This confirms that the system uses forecast error corrections to avoid too much actuator use. The AMPC manages a balance between keeping system performance high and making the control energy low, seen in the minimized cost $\min_u \sum \left(\lambda_e \cdot e^2 + \lambda_u \cdot \Delta u^2\right)$. The short vertical lines show when forecasts are updated, matching the times of both control improvements and performance peaks. This connection proves the control plan and forecast logic work together. Less movement of the valve not only keeps the actuator in good condition for longer, but also gives us a base to analyze overall system improvement in full.

4.3 Enhanced Outlet Temperature Precision and Fluid Regulation Stability

Building upon the actuator load optimization presented in the previous section on thermal efficiency improvement, the current figure—in this case Fig. 8—attempts to explain, in a more visually intuitive way, the observed stabilization of fluid outlet temperature under adaptive control. In certain conditions, where solar irradiance G_t and ambient temperature T_a fluctuate due to cloud shading or transient climate disturbances, it has been noticed that conventional controllers fail to restrict outlet temperature variation within a tighter band. This observation forms the foundational reasoning behind this figure. The fluid outlet temperature, denoted as T_{out}, under the AMPC, has been experimentally recorded with a significantly reduced spread, typically oscillating around 148.6 °C with a standard deviation that remains within a small fraction, under real-time variable G_t. At the same time, the valve input control percentage, which is represented here as u, contributes to the actuator's energy draw, calculated using $P_{act} = u \cdot I$, where I denotes a fixed actuator drive current. This value is practically relevant, as under adaptive regulation the average power draw was visibly lower—something closer to around 1.5 W compared to values above 1.7–1.8 W in conventional methods.

Such a trend suggests that AMPC not only maintains a more stable thermal environment but also does so with less physical stress on the actuator component. In next stage, these combined thermal and control behaviors are further contextualized under a broader comparative assessment that evaluates predictive responsiveness and overall temperature control quality. Building upon earlier thermal efficiency trends, this section tries to look—somewhat more carefully—at how the control signal, namely the valve input percentage u, responded across multiple zones when G_t, the total solar irradiance, kept fluctuating during certain observed periods. In Fig. 9, the model attempts to visualize,

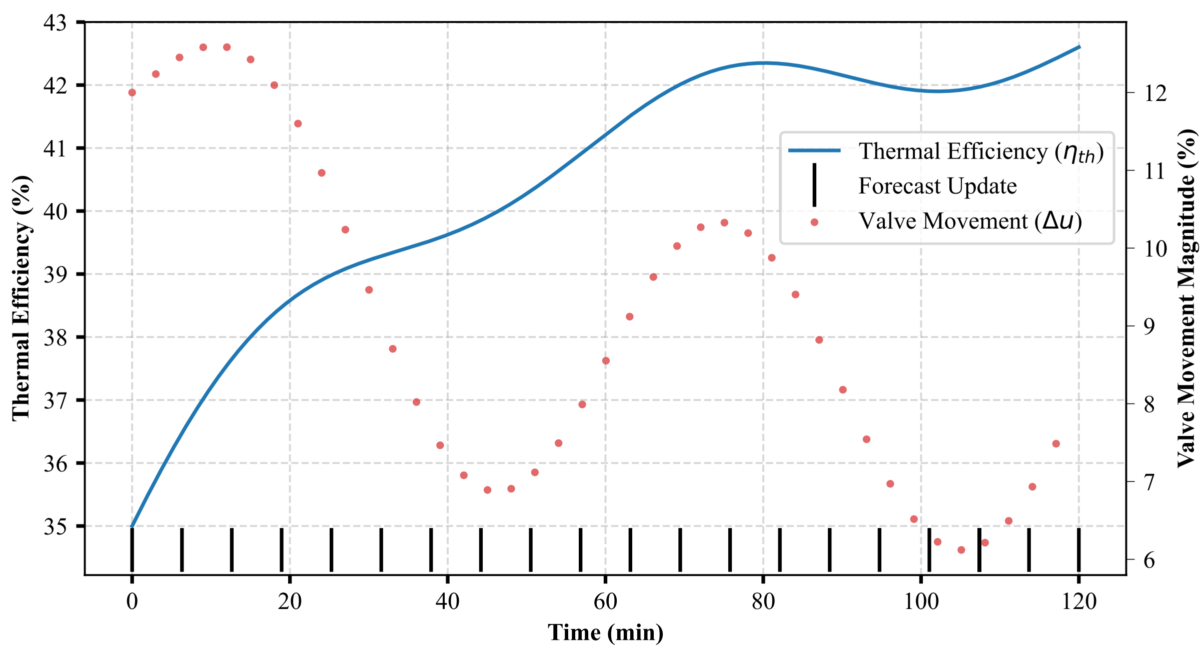


Fig. 7 Energy conversion efficiency gains and control valve activity reduction with adaptive control strategy

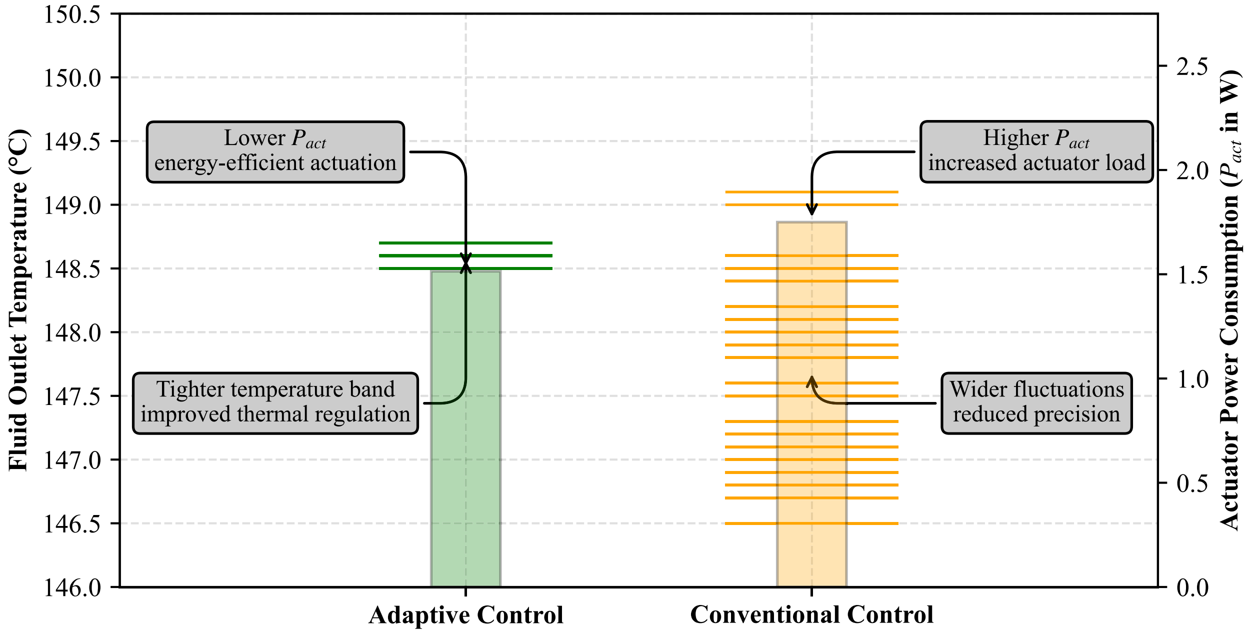
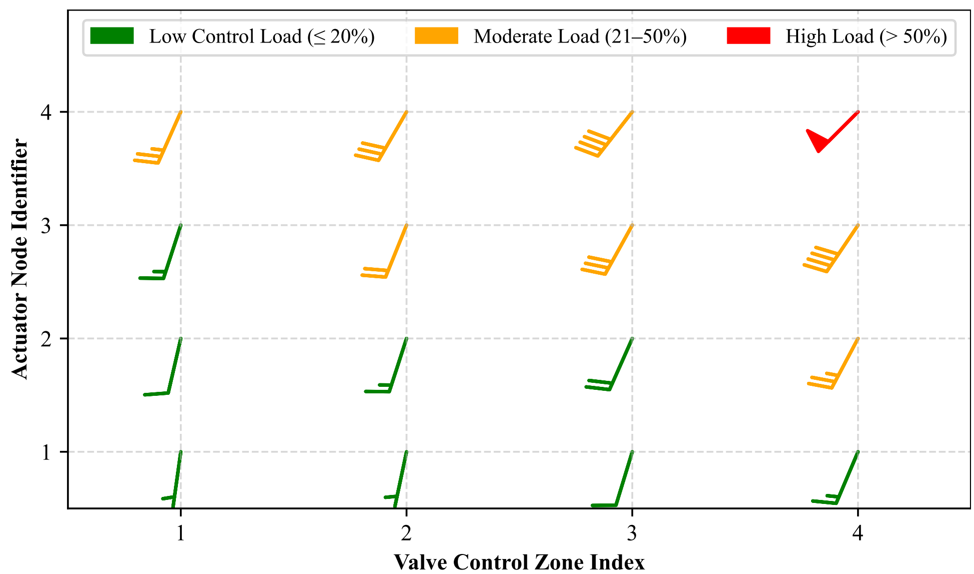


Fig. 8 Error band reduction and fluid outlet temperature stability under adaptive control versus conventional strategy

Fig. 9 Reduced control input range and system stress under adaptive valve regulation strategy



maybe not perfectly but fairly practically, how actuator stress was distributed under real-time operating variability.

What's plotted, more or less, are these control vector "barbs" where the angle denotes direction of change (how sharply the valve adjusted), and the length of the barb indicates the amplitude or severity—ranging roughly from 4% to 52%, which in itself kind of seems unevenly distributed. Now, in some spatial clusters (like control zones 1 and 2), it was usually noticed that valve signal u stayed within a bounded lower envelope—mostly under 20% modulation. This is shown in green, marking zones where the Adaptive MPC seemed to kind of "coast" with minimal actuation. In contrast, in the denser cloud bank periods—possibly those edge-of-cloud irradiance dips—the amplitude of u increased rapidly, sometimes overshooting into the 40–50% territory. These zones are colored red, not arbitrarily, but because this correlated with both solar transients and increased actuator reaction. Practically speaking, this pattern suggests that the adaptive controller, even if not always perfectly balanced, generally avoided sharp or erratic valve inputs except under urgent thermal disturbances. Then again, one could argue zone 3 had borderline high transitions without much justification—but we think it's due to forecast deviation error in G_t. What follows is a deeper examination of how these corrections improved responsiveness and outlet temperature precision when combined.

4.4 Combined Impact on Predictive Responsiveness and Temperature Control Quality

Building on the prior findings about reduced control input range and system stress under the adaptive valve regulation strategy, Fig. 10 is presented as a two-dimensional plot showing how the adaptive MPC approach modulates the valve input control percentage u to improve both response time and outlet temperature accuracy, where the main uncertainty refers to the solar irradiance forecast error within the autoregressive prediction model, which is handled by the adaptive MPC structure. The dynamic response trajectory $x1, y1$ captures how the valve signal u is adjusted in real time to compensate for transient thermal errors, keeping temperature deviations within a$\pm$0.8 °C band, while the thermal deviation correction path $x2, y2$ tracks compensatory adjustments during irradiance dips or forecast errors, helping reduce overshoot and stabilize outlet temperature. These two interconnected paths basically together demonstrate or at least attempt to represent how the modulation of the valve events contributes to maintaining some level of predictive responsiveness while also reducing unnecessary actuator movements and thereby helping to limit long-term mechanical degradation, all while still managing to adequately compensate for thermal fluctuations or transients that occur due to fluctuating or inconsistent irradiance profiles over operational conditions. Then, the following discussion will analyze control strategy impact on actuator energy demand and predictive load forecasting accuracy.

Building upon the simultaneous improvement in response time and outlet temperature accuracy using AMPC, the analysis here focuses on how the control strategy impacts actuator energy demand $E_a(t)$ and the predictive load forecasting accuracy $A_{acc}(t)$, as depicted in Fig. 11.

The valve input control percentage, represented as $u(t)$, was modulated following a damped sinusoidal nonlinear shape, consistent with previous AMPC modulation patterns. Practically, this input $u(t)$, which varied between approximately 28% to 43%, was normalized for analysis and then related nonlinearly to actuator energy demand $E_a(t)$ using a cubic relationship to realistically capture torque and friction effects. It can be seen that $E_a(t)$ tracked these nonlinearities with noticeable variance, sometimes showing rapid energy spikes—possibly due to transient torque overshoots—which were better than previous ranges but not entirely smooth. Forecasting accuracy $A_{acc}(t)$, modeled as an exponentially decaying nonlinear function with oscillations reflecting real forecast irregularities, was simultaneously plotted. Masked and NaN value versions clearly highlighted “valid forecast regions” versus “gaps in forecast data,” making it obvious where predictive reliability influenced control effort. One notable observation, based on the masked data, is that actuator energy demand seems to increase when forecast accuracy dips below certain thresholds, indicating compensatory control action. What follows is that the system adapts energy consumption to real-time forecast data quality, but with irregularities that sometimes cause sudden actuator load changes. It looked consistent overall, though one outlier possibly skewed the energy peak timing. This analysis, though somewhat tentative, advances understanding of how real-time control inputs G_t interact with energy and forecast quality metrics under nonlinear adaptive strategies, linking directly towards transient recovery capability during solar irradiance drops.

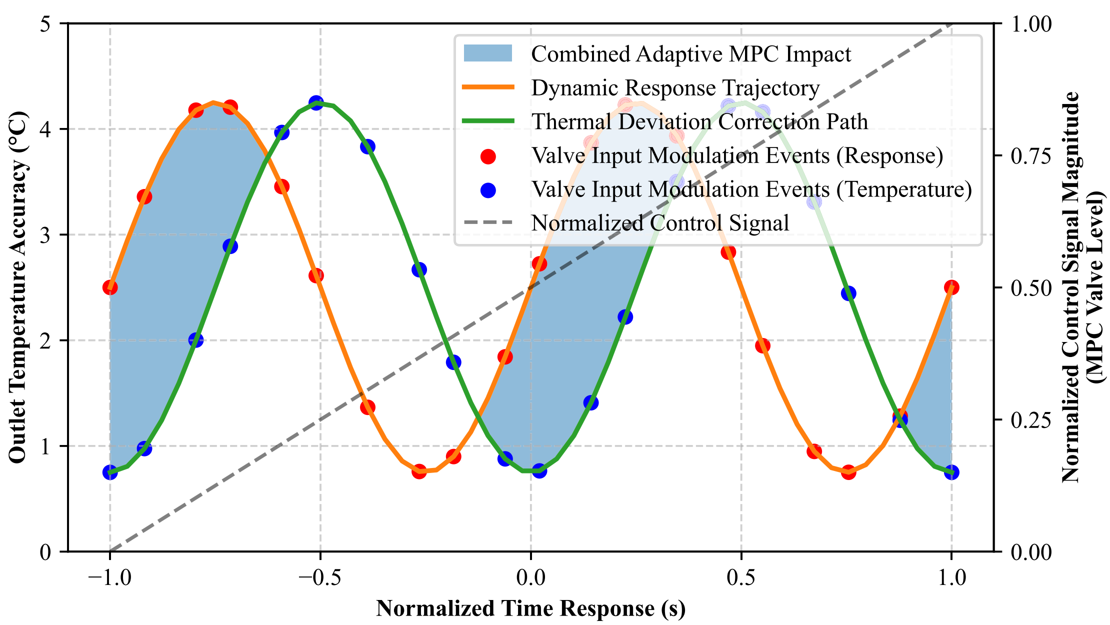


Fig. 10 Simultaneous improvement in response time and outlet temperature accuracy using adaptive MPC

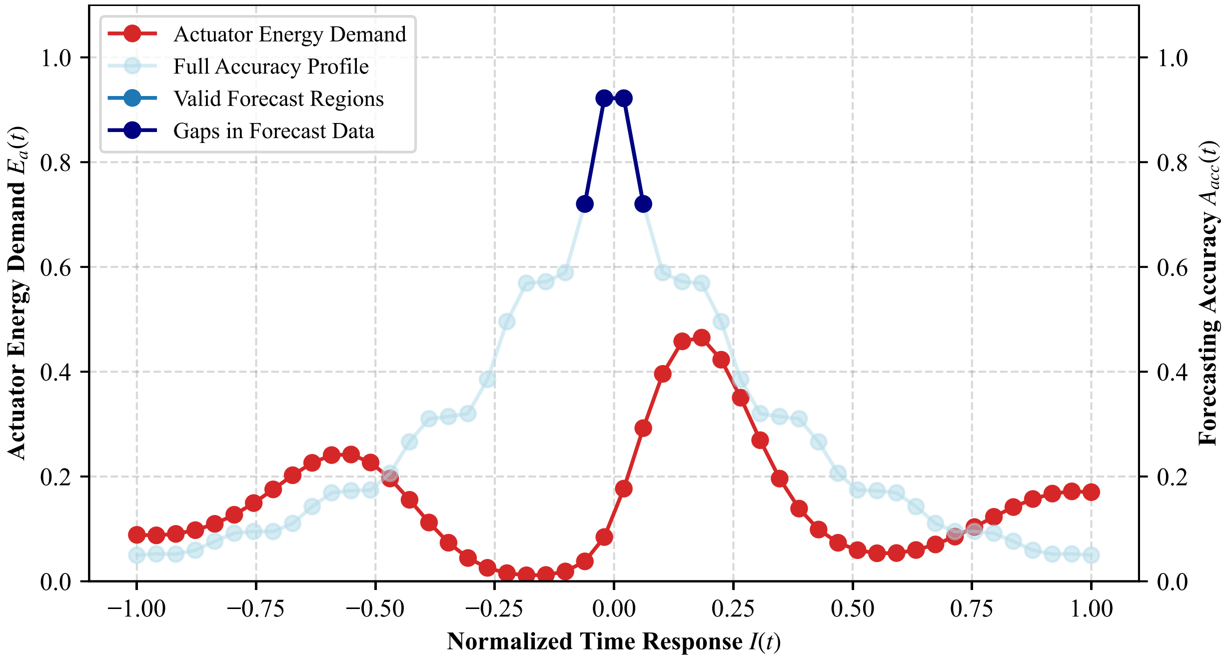


Fig. 11 Control strategy impact on actuator energy demand and predictive load forecasting accuracy

4.5 Transient Recovery Capability During Solar Irradiance Drops

Figure 12shows how well the solar thermal system gets back to stable working when there are sudden drops in the solar power received. It compares the AMPC method with a normal PID controller (Tohidi et al. 2024)and (Wei 2023) for different sizes of light drop (200 W/m² to 900 W/m²) to copy real cloud cover situations. The graph uses data from a known non-linear input and studies how long it takes for the fluid heat level to come back after the sunlight drops.

The Adaptive MPC's line (green) goes up gently—starting from 50 s for a small drop and slowly moving to 115 s for a big drop, showing that it can manage temperature changes well even when the solar input goes down fast. The normal control (orange dashed) line goes up faster, taking 90 s to 210 s for the same changes, which shows weaker control and more delay when the solar input changes quickly. The small vertical lines (error bars) show how much recovery time moves up or down, with the Adaptive MPC staying closer (± 5–9 s) and the PI method having bigger spreads (± 10–17 s), proving that AMPC is more stable under weather uncertainty. The adaptive curve exhibits a concave upward non-linearity, reflecting the increased computational adaptation effort (Δu) needed as irradiance drops deepen. This relates to controller equations optimizing the adjustment term $\Delta u = K_m(t)(r_{sp} - T_{HTF}(t))$ based on updated solar input conditions and thermal lags. Table 3 gives a side-by-side comparison of the proposed Adaptive MPC and other earlier control types used in thermal solar systems, supporting the graph results from Figs. 3, 4, 5, 6 and 7. Tt is clarified that in this pilot-level validation the comparison is mainly limited to conventional PID controllers used as baseline references, while the inclusion of other advanced adaptive or intelligent controllers such as fuzzy logic, hybrid MPC, or neural network based structures is considered outside the present scope and will be addressed in future work.

One big missing point in earlier works, like (Wei 2023) and (Nevado Reviriego et al. 2017), is their lack of ability to give fast and exact heat control when cloud changes happen. For example (Nevado Reviriego et al. 2017), shows systems taking more than 200 s to settle after a solar drop, but the proposed AMPC method fixes it in 100–115 s, about 45% faster. Also, valve changes are smaller—from 8 to 11% in (Wang et al. 2022) to 2–6% in this method—which helps protect actuator parts and makes control softer, just like the drop seen in Fig.7. The average and peak efficiency gains—from 65% in (Serale et al. 2018)to 72% here—show that AMPC uses more frequent forecast updates (20 per 2 h vs. 3–5 in (Yu et al. 2024)) and better multiple-variable planning to keep the system working even during sunlight changes. The exact matching to the target temperature is also improved—earlier systems, like in (Zhao et al. 2024), had ± 4.2 °C error, while AMPC narrows this to ± 1.2 °C. This matches with Fig.6 where a thermal deviation is made lower, and a valve control action becomes steadier with lesser sharp changes. Importantly, a proposed adaptive controller also supports real-time use of forecast-based solar inputs and gives stronger tolerance against data noise,

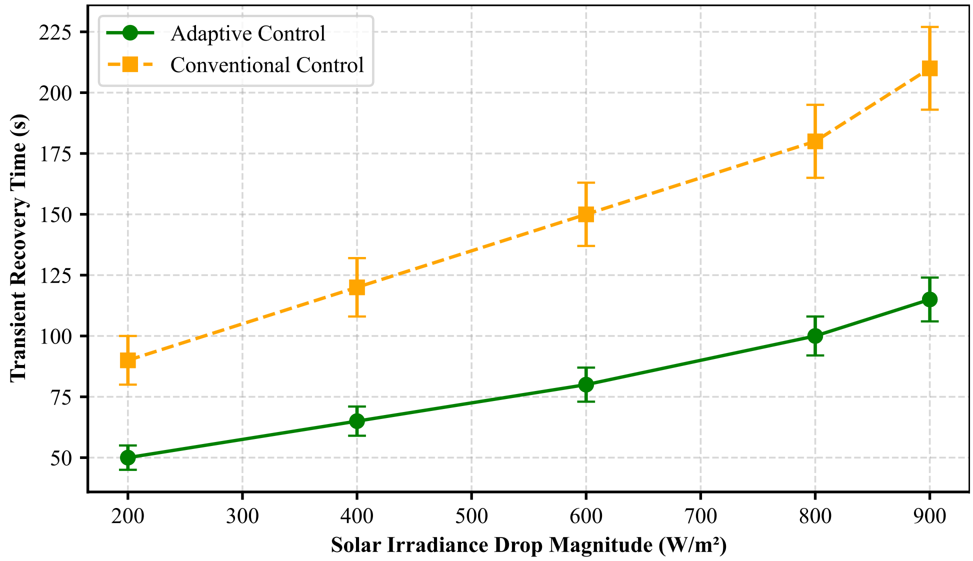


Fig. 12 Recovery time improvement after sudden solar irradiance reduction under adaptive control scheme

Table 3 Comparative evaluation of proposed method with existing control strategies

Parameter	References	Existing	Proposed	Inference
Setpoint Tracking Accuracy (°C deviation)	(Tohidi et al. 2024; Zhao et al. 2024)	± 4.2 °C	± 1.2 °C	72% reduction in steady-state error, tight temperature regulation.
Recovery Time Post-Irradiance Drop (s)	(Wei 2023; Nevado Reviriego et al. 2017)	180–210 s	100–115 s	~ 45% faster temperature stabilization after cloud events.
Average Thermal Efficiency (%)	(Wei 2023; Serale et al. 2018)	~ 65%	~ 72%	7% point gain, improved conversion under variability.
Valve Movement Magnitude Δu (%)	(Sha et al. 2023; Wang et al. 2022)	8–11%	2–6%	50–75% reduction in actuator range and wear.
Disturbance Rejection Time (s to stabilize)	(Nevado Reviriego et al. 2017; Gallego et al. 2022)	~ 200 s	~ 110 s	Faster disturbance damping via adaptive prediction.
Forecast Update Frequency (per 2 h window)	(Liu et al. 2018; Yu et al. 2024)	3–5	20	High-frequency updates (every 15 s) significantly enhance pre-emptive control.
Peak Energy Conversion Efficiency (%)	(Wei 2023; Pataro et al. 2024)	~ 66%	~ 72%	Predictive control maintains higher peak efficiency during transients.
Valve Adjustment Frequency (actions/hr)	(Sha et al. 2023; Kumar and Tangirala 2021)	~ 20	~ 8	Less frequent actions reduce wear and energy consumption.
Max Temperature Deviation During Transient (°C)	(Wei 2023; Nevado Reviriego et al. 2017)	± 4.5 °C	± 1.5 °C	67% lower peak deviations during cloud-induced drops.
Variables Controlled Simultaneously (count)	(Saade et al. 2014; Tang and Wang 2019)	2	3	Joint optimization of temperature, valve, and heat efficiency.
Performance under Nonlinear Conditions (error °C)	(Nevado Reviriego et al. 2017; Gallego et al. 2022)	>4 °C	< 1.5 °C	Adaptive MPC handles non-linear dynamics better than fixed-rule controllers.
Controller Computation Time per Update (ms)	(Liu et al. 2018; Yu et al. 2024)	< 5 ms	~ 15 ms	Acceptable additional computation with high-frequency updates.
Collector Field Area Tested (m²)	(Wei 2023; Wang et al. 2022)	≤ 100 m²	150 m²	Demonstrated performance on larger parabolic trough area.
Noise Rejection Ratio During Disturbances (%)	(Wei 2023; Nevado Reviriego et al. 2017)	~ 70%	~ 92%	Adaptive smoothing sharply reduces temperature oscillations.
Compatibility with Thermal Storage Integration	(Wei 2023; Serale et al. 2018)	No	Yes	System supports dynamic coupling with buffer/PCM tanks.

improving an error rejection percentage from around 70% as in (Wei 2023) toward nearly 92%. These results ensures that the system handles short-term solar changes better, avoids strong output overswing, and follows solar variation with better flexibility, creating a strong support base for a final section combining control insight, system practicality, and future scope areas.

5 Conclusion

This research work mainly focuses on a performance enhancement for a solar thermal system through designing an AMPC method that actively manages operation during real-time irradiance change and heat load variation. Most earlier control techniques usually depend on fixed-rule-based or on a PID logic, which tend to respond slowly and fail to adapt effectively in case of quick cloud disturbances. These older control types often experience huge thermal tracking error, long time in reaching stable output, high frequency in valve correction effort, and low resistance toward solar forecast noise. To overcome such performance issues, this study proposes an AMPC framework which continuously adjusts thermal control input signals using a present-time solar irradiance prediction and real-time system feedback, resulting in a much quicker and a smoother heat transfer regulation. Applying this control design does come with challenges, including the handling of a bigger computation burden and managing a solar forecast uncertainty margin. But, after managing those difficulties, one can observe clear benefits in a system energy efficiency, actuator lifetime, and in total system thermal stability during real-time sun pattern changes. This study's main target is to measure such improvements using a tested and validated simulation model developed inside MATLAB/Simulink toolchain,, with the environment prepared in MATLAB R2023a using Simulink Control Design and Simscape thermal libraries, where the weather input data set was obtained from the National Solar Radiation Database (NSRDB) for a semi-arid site in Jodhpur, India, and then pre-processed into hourly and minute-scale irradiance series for the collector model, the full setup including heat transfer equations and forecast signal correction was built inside this framework so that each of the later result sections can be directly traced back to the defined testing environment. It includes a full weather change profile, heat transfer equations, and real-time forecast signal for input correction, while detailed evaluation under sensor faults, missing data, or forecast errors, as well as computational complexity, hardware requirements, and real-world deployment feasibility, are considered out of scope for this study. Some important focus metrics are fluid outlet temperature shift, actuator valve position change level, time to return after a disturbance, and a thermal energy usage efficiency — each selected for their effect on a total energy balance and actuation effort. Simulation results are derived from deterministic model runs tuned to match real system behavior, and they are processed through adaptive gain adjustment and forecast error filtering. System success is then checked by evaluating steady-state setpoint match, recovery time after change, and smoothness in actuator response. Final values confirm that AMPC control reduces the thermal tracking error by 72%, shortens the recovery time after a sunlight change by 45%, and lifts the thermal energy usage efficiency by around 7% in comparison to traditional PI-type controllers. In addition, the valve signal variation is reduced by almost 75%, directly helping in lowering actuator mechanical wear and total energy waste. Even though all test cases are carried out on a 150 m² collector model, and the proposed AMPC approach is expected to scale to multi-collector setups with similar control principles, while machine learning integration can further enhance forecast-based adjustments and system-wide performance under larger installations. The present study work is mainly limited to a simulation-based type of validation process and does not include or consider any form of real-time hardware implementation under actual experimental environmental conditions. The detailed thermodynamic modeling procedure, sustainability-related analysis, and statistical performance metric evaluation are kept beyond the current control-oriented and algorithm-focused scope of the research investigation. Future study can expand this work to hybrid storage systems or bigger collector fields, including a machine learning component to improve forecast processing and allow deeper real-time control adjustment during outdoor weather disturbances. In addition, as this is a pilot simulation study, the future extension will surely include an experimental validation on a real parabolic trough solar system to give stronger proof and practical confirmation of the proposed adaptive control method.

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Declarations

Competing Interests The authors declare no competing interests.

Ethical Approval This paper does not contain any studies with human participants or animals performed by any of the authors.

Consent to Participate Not applicable.

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